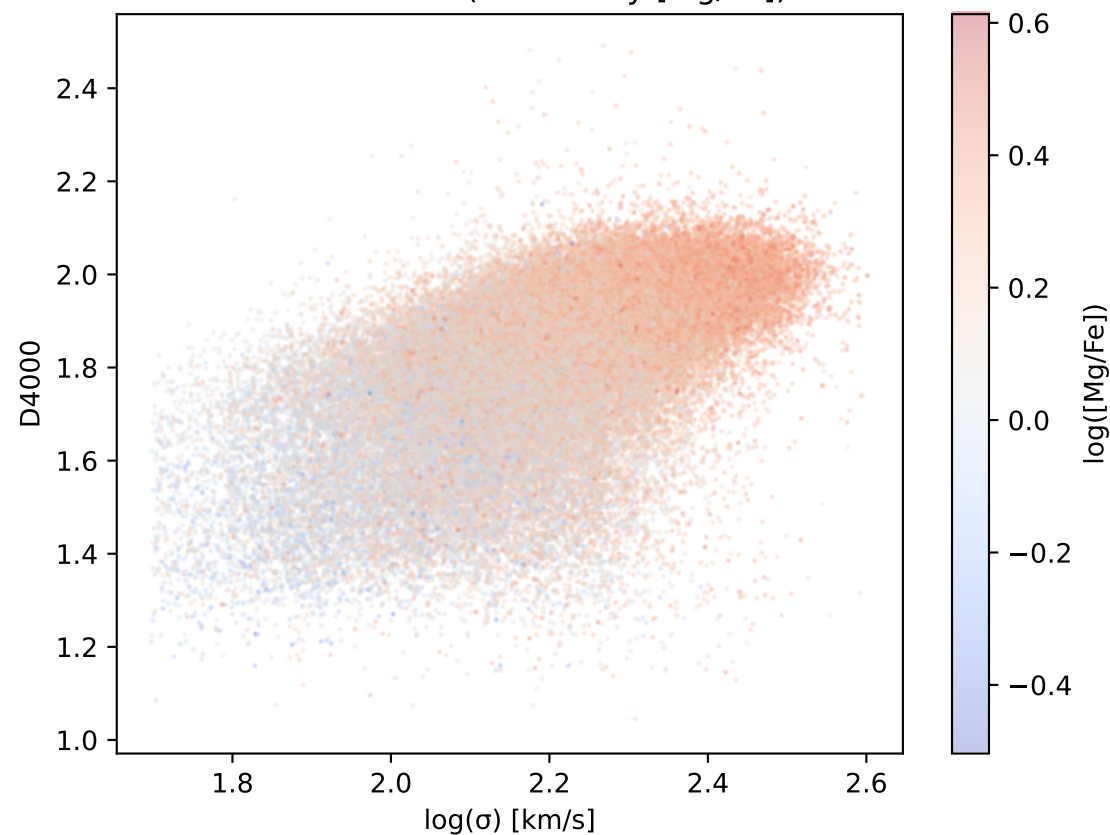


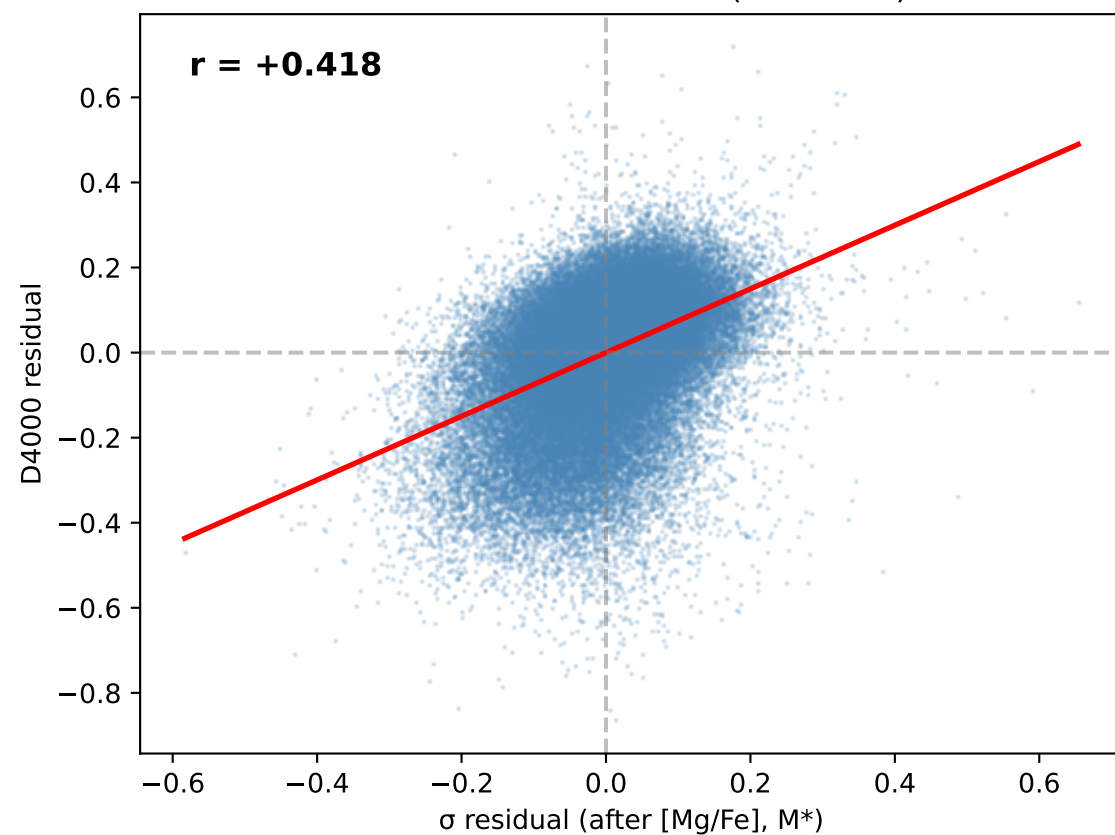
Chemical Clock vs Spectroscopic Clock Test

TEP Prediction: At fixed [Mg/Fe], high- σ galaxies appear YOUNGER (lower D4000)

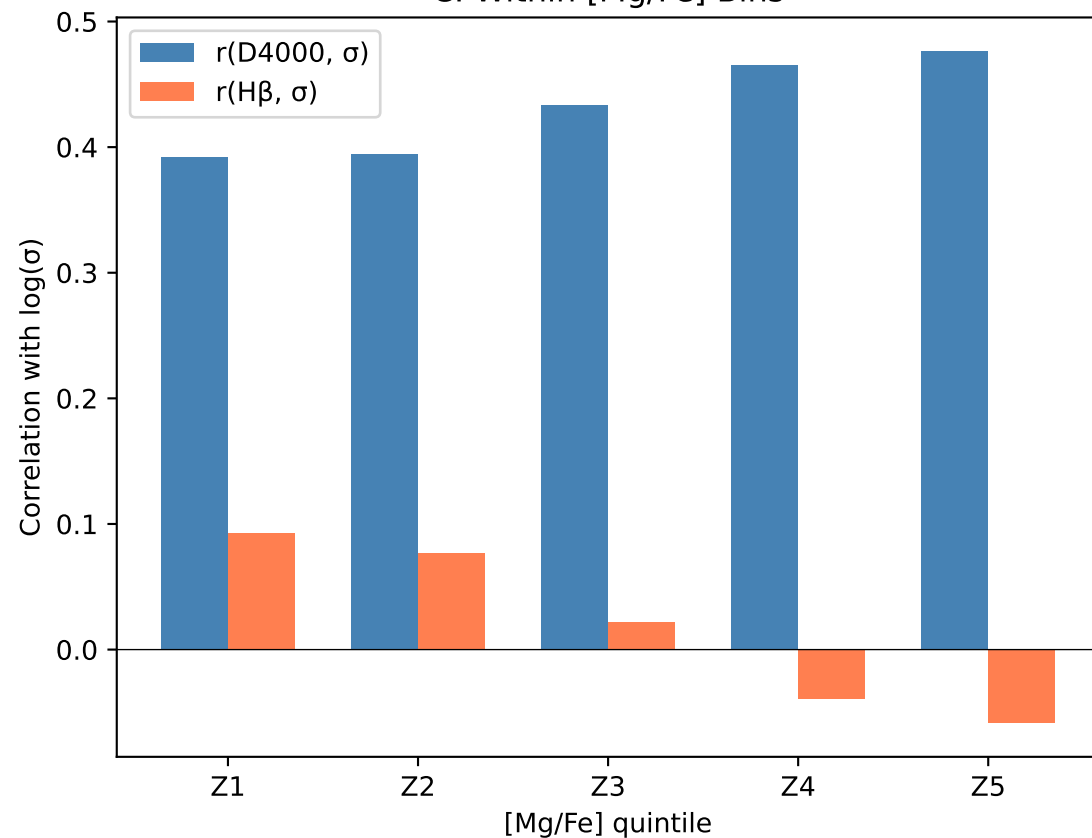
A. D4000 vs σ (colored by [Mg/Fe])



B. Residual Correlation (THE TEST)



C. Within [Mg/Fe] Bins



D. Matched-Pair Summary

Matched-pair cells: 427

Weighted $r(D4000, \sigma) = +0.4466$

Cells with $r < 0$: 2 (0.5%)

Sign test $p = 0.0000$

STANDARD PHYSICS